

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Panasonic
Firmware Version: N/A
Model(s): PT-VMZ51S, PT-BMZ51, PT-VMZ41, PT-VMZ71, PT-VMZ61, PT-VMW61, PT-VMW51

Tested on the Following Software and Firmware Versions

Global Scriptor Version	2.23.0.13
IP Link Pro Control Processor Firmware	3.18.0001-b002
IP Link Pro Control Processor xi Firmware	1.10.0001-b002
IP Link Pro Control Processor Q xi Firmware	1.10.0002-b002

Version History

Module Version	Date	Notes
1_0_3_0	3/20/2025	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword must be set accordingly.
Example: `InterfaceName.devicePassword = 'extron'`
- For Serial control, DeviceID variable must be set accordingly. Default value is '1'. DeviceID ranges from '1' to '64', and Broadcast. If Device ID is set to 'Broadcast', status is unavailable.
Example: `InterfaceName.DeviceID = '1'`
- For Serial control, a command pacing of 0.5 seconds is recommended between all commands.

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='PT-VMZ51S')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='PT-VMZ51S')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 4352, Model='PT-VMZ51S')</code>

Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Auto' 'Native' 'V-Fit'	Value 'Normal' 'Full'	Value 'Wide' 'H-Fit'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Auto')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'Computer 1' 'HDMI 2' 'Digital Link' ¹	Value 'Computer 2' 'Network/USB'	Value 'HDMI 1' 'Memory Viewer'
# Input example InterfaceName.Set('Input', 'Computer 1')			
Command Keypad	Value '1' – '6'		
# Keypad example InterfaceName.Set('Keypad', '1')			
Command LampMode	Value 'Normal' 'User'	Value 'Eco'	Value 'Quiet'
# LampMode example InterfaceName.Set('LampMode', 'Normal')			
Command MenuNavigation	Value 'Menu' 'Up' 'Right'	Value 'Return' 'Down'	Value 'Enter' 'Left'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			
Command PictureMode	Value 'Dynamic' 'White Board'	Value 'Natural' 'Cinema'	Value 'Standard' 'Dicom Sim.'
# PictureMode example InterfaceName.Set('PictureMode', 'Dynamic')			

Global Scripter Module Communication Sheet

Command	Value	Value
Power	'On'	'Off'
# Power example InterfaceName.Set('Power', 'On')		
Command	Value	Value
Shutter	'On'	'Off'
# Shutter example InterfaceName.Set('Shutter', 'On')		
Command	Value	
Volume	0 to 63 in steps of 1	
# Volume example InterfaceName.Set('Volume', 63)		

¹ Only available for the PT-VMZ71, PT-VMZ61, PT-VMW61, PT-VMZ51, PT-VMW51 models

Status Available (Serial)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value 'Auto' 'Native' 'V-Fit'	Value 'Normal' 'Full'	Value 'Wide' 'H-Fit'
# AspectRatio example InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command FilterUsage	Value Hours		
# FilterUsage example InterfaceName.Update('FilterUsage') Value = InterfaceName.ReadStatus('FilterUsage') InterfaceName.SubscribeStatus('FilterUsage', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'Computer 1' 'HDMI 2' 'Digital Link' ¹	Value 'Computer 2' 'Network/USB'	Value 'HDMI 1' 'Memory Viewer'
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			

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Command LampMode	Value 'Normal' 'User'	Value 'Eco'	Value 'Quiet'
# LampMode example InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command LampUsage	Value Hours		
# LampUsage example InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command OperationHours	Value Hours		
# OperationHours example InterfaceName.Update('OperationHours') Value = InterfaceName.ReadStatus('OperationHours') InterfaceName.SubscribeStatus('OperationHours', None, FeedbackHandler)			
Command PictureMode	Value 'Dynamic' 'White Board'	Value 'Natural' 'Cinema'	Value 'Standard' 'Dicom Sim.'
# PictureMode example InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command Shutter	Value 'On'	Value 'Off'	
# Shutter example InterfaceName.Update('Shutter') Value = InterfaceName.ReadStatus('Shutter') InterfaceName.SubscribeStatus('Shutter', None, FeedbackHandler)			
Command Volume	Value 0 to 63 in steps of 1		
# Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)			

¹ Only available for the PT-VMZ71, PT-VMZ61, PT-VMW61, PT-VMZ51, PT-VMW51 models

Control Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	
AudioMute	'On'	'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command	Value	Value	
AVMute	'On'	'Off'	
# AVMute example InterfaceName.Set('AVMute', 'On')			
Command	Value	Value	
Freeze	'On'	'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command	Value	Value	Value
Input	'Computer 1'	'Computer 2'	'HDMI 1'
	'HDMI 2'	'Memory Viewer'	'Network'
	'Digital Link' ¹		
# Input example InterfaceName.Set('Input', 'Computer 1')			
Command	Value	Value	Value
LampMode	'Normal'	'Eco'	'Quiet'
	'User'		
# LampMode example InterfaceName.Set('LampMode', 'Normal')			
Command	Value	Value	
Power	'On'	'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command	Value	Value	
VolumeStep	'Up'	'Down'	
# VolumeStep example InterfaceName.Set('VolumeStep', 'Up')			

¹ Only available for the PT-VMZ71, PT-VMZ61, PT-VMW61, PT-VMZ51, PT-VMW51 models

Status Available (Ethernet)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AudioMute	'On'	'Off'	
# AudioMute example InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command	Value	Value	Value
AVMute	'On'	'Off'	
# AVMute example InterfaceName.Update('AVMute') Value = InterfaceName.ReadStatus('AVMute') InterfaceName.SubscribeStatus('AVMute', None, FeedbackHandler)			
Command	Value	Value	Value
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	Value
DeviceStatus	'Normal'	'Multiple Warnings/Errors'	'Fan Warning'
	'Fan Error'	'Lamp Warning'	'Temperature Warning'
	'Temperature Error'	'Filter Warning'	'Filter Error'
	'Other Warning'	'Other Error'	
# DeviceStatus example InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command	Value	Value	Value
FilterUsage	Hours		
# FilterUsage example InterfaceName.Update('FilterUsage') Value = InterfaceName.ReadStatus('FilterUsage') InterfaceName.SubscribeStatus('FilterUsage', None, FeedbackHandler)			
Command	Value	Value	Value
Freeze	'On'	'Off'	
# Freeze example InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			

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Command Input	Value 'Computer 1' 'HDMI 2' 'Digital Link' ¹	Value 'Computer 2' 'Memory Viewer'	Value 'HDMI 1' 'Network'
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampMode	Value 'Normal' 'User'	Value 'Eco'	Value 'Quiet'
# LampMode example InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command LampUsage	Value Hours		
# LampUsage example InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	Value 'Cooling Down'
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			

¹ Only available for the PT-VMZ71, PT-VMZ61, PT-VMW61, PT-VMZ51, PT-VMW51 models

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

To ensure that the module functions correctly, please set the following:

- Menu → PROJECTOR SETUP → ECO MANAGEMENT → STANDBY MODE: NORMAL.
- Menu → PROJECTOR SETUP → RS-232C → PARITY: NONE.
- Menu → PROJECTOR SETUP → RS-232C → EMULATE: OFF.
- Menu → PROJECTOR SETUP → RS-232C → INPUT SELECT: PROJECTOR or DIGITAL LINK, depending on whether communication occurs through the projector or the optional Digital Link output supported device.
 - When communicating through Digital Link, ensure that Menu → NETWORK → DIGITAL LINK MODE is not set to ETHERNET.

Serial communication

Port Type: RS-232

Baud Rate: 9600

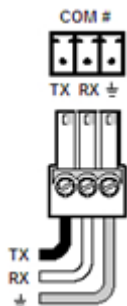
Data Bits: 8

Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Network Communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet (TCP)

Default Port: 4352

Logon Credentials Supported: Yes

Default Username: dispadmin

Default Password: None

Multi-Connection Capabilities: Undetermined

Port Changeability: No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

To ensure the functionality of the module, please set the following menu options:

- Menu → PROJECTOR SETUP → ECO MANAGEMENT → STANDBY MODE: NORMAL
- Menu → NETWORK → NETWORK CONTROL → PJLink CONTROL: ON

The password used for Ethernet control is the same password set for web control. If you wish to use Ethernet control without a password, disable the password for web control.

Appendix A. Set Commands(Serial)

Aspect Ratio Auto	\x02AD01;VSE:0\x03
Aspect Ratio Full	\x02AD01;VSE:6\x03
Aspect Ratio H-Fit	\x02AD01;VSE:9\x03
Aspect Ratio Native	\x02AD01;VSE:5\x03
Aspect Ratio Normal	\x02AD01;VSE:1\x03
Aspect Ratio V-Fit	\x02AD01;VSE:10\x03
Aspect Ratio Wide	\x02AD01;VSE:2\x03
Audio Mute Off	\x02AD01;AMT:0\x03
Audio Mute On	\x02AD01;AMT:1\x03
Auto Image None	\x02AD01;OAS\x03
Freeze Off	\x02AD01;OFZ:0\x03
Freeze On	\x02AD01;OFZ:1\x03
Input Computer 1	\x02AD01;IIS:RG1\x03
Input Computer 2	\x02AD01;IIS:RG2\x03
Input Digital Link	\x02AD01;IIS:DL1\x03
Input HDMI 1	\x02AD01;IIS:HD1\x03
Input HDMI 2	\x02AD01;IIS:HD2\x03
Input Memory Viewer	\x02AD01;IIS:MV1\x03
Input Network/USB	\x02AD01;IIS:NWP\x03
Keypad 1	\x02AD01;ONK:1\x03
Keypad 6	\x02AD01;ONK:6\x03
Lamp Mode Eco	\x02AD01;VXX:LPWI1=+00001\x03
Lamp Mode Normal	\x02AD01;VXX:LPWI1=+00000\x03
Lamp Mode Quiet	\x02AD01;VXX:LPWI1=+00040\x03
Lamp Mode User	\x02AD01;VXX:LPWI1=+00100\x03
Menu Navigation Down	\x02AD01;OCD\x03
Menu Navigation Enter	\x02AD01;OEN\x03
Menu Navigation Left	\x02AD01;OCL\x03
Menu Navigation Menu	\x02AD01;OMN\x03
Menu Navigation Return	\x02AD01;OBK\x03
Menu Navigation Right	\x02AD01;OCR\x03
Menu Navigation Up	\x02AD01;OCU\x03
Picture Mode Cinema	\x02AD01;VPM:CIN\x03
Picture Mode Dicom Sim.	\x02AD01;VPM:DIC\x03
Picture Mode Dynamic	\x02AD01;VPM:DYN\x03
Picture Mode Natural	\x02AD01;VPM:NAT\x03
Picture Mode Standard	\x02AD01;VPM:STD\x03
Picture Mode White Board	\x02AD01;VPM:WBD\x03
Power Off	\x02AD01;POF\x03
Power On	\x02AD01;PON\x03
Shutter Off	\x02AD01;OSH:0\x03
Shutter On	\x02AD01;OSH:1\x03

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Volume 0	\x02AD01;AVL:000\x03
Volume 63	\x02AD01;AVL:063\x03

Appendix B. Update Commands(Serial)

Aspect Ratio	\x02AD01;QS1\x03
Audio Mute	\x02AD01;QMT\x03
Filter Usage	\x02AD01;QFI:0\x03
Freeze	\x02AD01;QFZ\x03
Input	\x02AD01;QIN\x03
Lamp Mode	\x02AD01;QLP\x03
Lamp Usage	\x02AD01;Q\$L\x03
Operation Hours	\x02AD01;QVX:RTMI0\x03
Picture Mode	\x02AD01;QPM\x03
Power	\x02AD01;QPW\x03
Shutter	\x02AD01;QSH\x03
Volume	\x02AD01;QAV\x03

Appendix C. Set Commands(Ethernet)

Audio Mute Off	%1AVMT 20\x0D
Audio Mute On	%1AVMT 21\x0D
AV Mute Off	%1AVMT 30\x0D
AV Mute On	%1AVMT 31\x0D
Freeze Off	%2FREZ 0\x0D
Freeze On	%2FREZ 1\x0D
Input Computer 1	%1INPT 11\x0D
Input Computer 2	%1INPT 12\x0D
Input HDMI 1	%1INPT 31\x0D
Input HDMI 2	%1INPT 32\x0D
Input Memory Viewer	%1INPT 41\x0D
Input Network	%1INPT 51\x0D
Power Off	%1POWR 0\x0D
Power On	%1POWR 1\x0D
Volume Step Down	%2SVOL 0\x0D
Volume Step Up	%2SVOL 1\x0D

Appendix D. Update Commands(Ethernet)

Audio Mute	%1AVMT ?\x0D
Device Status	%1ERST ?\x0D
Filter Usage	%2FILT ?\x0D
Freeze	%2FREZ ?\x0D
Input	%1INPT ?\x0D
Lamp Usage	%1LAMP ?\x0D
Power	%1POWR ?\x0D